

Gold-Free Faithfulness Metrics for NL-to-FOL Translation

Internal Research Report

AI Inventor Research

September 2026

Contents

1	Framing	4
2	Iteration 1	4
2.1	Strategy	4
2.2	Experiment A: FOL-Triage	5
2.2.1	Headline AUROCs (track L, ERROR vs CORRECT, $n = 477$, 199 errors) . .	5
2.2.2	L1 lint: a negative result on LLM outputs	5
2.2.3	L2 role-aware check: below bag-of-words	6
2.2.4	Decomposed judge ablation	6
2.2.5	Per-type sensitivity (track L, at binary threshold)	6
2.2.6	Per-type sensitivity (track H)	7
2.2.7	Invariance under meaning-preserving rewrites	7
2.2.8	Coverage	7
2.2.9	Cost	9
2.2.10	Contamination check	9
2.2.11	Complexity strata (track L, fused score)	9
2.2.12	Per-system AUROC (descriptive)	9
2.3	Experiment C: Cross-System Consensus	10
2.3.1	Headline AUROCs (track L, ERROR vs CORRECT, $n = 546$, 249 errors) . .	10
2.3.2	RENAME rewrite invariance: failure	10
2.3.3	Blind-spot analysis	10
2.3.4	Cost	10
2.4	Experiment D: Disguised Judge and Baselines	12
2.4.1	Pilot structural metrics: negative result	12
2.4.2	Round-trip verification	12
2.4.3	Cross-fitted baseline combinations (track L)	14
2.4.4	Contamination analysis	14
2.4.5	Invariance of judges under meaning-preserving rewrites	14
2.4.6	Per-error-type sensitivity of judges (track H)	15
2.4.7	Complexity analysis (track L)	15
2.5	Iteration 1 Summary	15

3	Iteration 2	17
3.1	Strategy	17
3.2	Experiment 5: PEER+TEXT on development dataset E	17
3.2.1	Name-free alignment: failed the pre-registered screen gate	17
3.2.2	Headline AUROCs on E (R_AB, ERROR vs CORRECT)	19
3.2.3	Does fusion beat <code>c_score_align</code> alone?	19
3.2.4	Mechanism predictions	19
3.2.5	Rewrite false alarms (screen, CORRECT bases)	19
3.2.6	System-level correlation	19
3.3	Evaluation 1: Re-evaluation under alternative label regimes	19
3.3.1	Label regime shifts	20
3.4	Dataset 3: R_COMP and PERTURB	20
3.5	Experiment 6: Baselines on development set E	20
3.6	Dead ends: B2 and R_ADJ	22
3.7	Iteration 2 Summary	22
4	Iteration 3	22
4.1	Strategy	22
4.2	Test T1: Head-on API bar on development set E	23
4.2.1	Verdict	23
4.2.2	Item-level AUROC on R_AB (E_POOL PRIMARY)	23
4.2.3	Paired deltas: challenger minus bar (stratified)	23
4.2.4	Cross-fitted stacks and nesting	24
4.2.5	VEX confound control	24
4.2.6	AUROC by complexity bin	24
4.2.7	Recall per error group at matched FA 0.10	25
4.2.8	System-level correlation	25
4.3	Test T2: R_COMP (long composed sentences)	25
4.3.1	SIG: primary test	25
4.3.2	SIG: per-metric AUROCs	26
4.3.3	Rename invariance on R_COMP	26
4.3.4	FREE condition: NOT_TESTABLE	26
4.4	Tests T3 + T5: PERTURB sensitivity and rename tradeoff	26
4.4.1	T3: Per-operator within-base AUROC	26
4.4.2	T5: Rename tradeoff table	27
4.5	Test T4: Mechanism verification	27
4.6	Prior-art positioning	28
4.7	Iteration 3 Summary	28
5	Iteration 4	29
5.1	Strategy	29
5.2	Experiment 9: CSC on development set E — provisional negative result	29
5.2.1	Anchoring: the core finding	29
5.2.2	Headline AUROCs on E	29
5.2.3	Anchoring by error class	31
5.3	Experiment 10: CSC on PERTURB — budget exhausted	31
5.3.1	SIGPROXY: cued consensus on R_COMP	31
5.3.2	Same-vocabulary typed repair	31

5.4	Dataset 4: E2 confirmation set — partial	32
5.5	Dataset 5: R_COMP FREE name-free labels	32
5.5.1	Executor audit	32
5.6	Evaluation 3: T8 vocabulary analysis	32
5.7	Iteration 4 Summary	33
5.8	Dead ends accumulated	33
6	Iteration 5	33
6.1	Strategy	33
6.2	Experiment 11: E2-A confirmation	33
6.2.1	DT stratum (testable)	34
6.2.2	Long pool (L25 + EXC, not testable)	34
6.2.3	ALL cell	34
6.3	Experiment 12: E2-B (long-sentence pool)	34
6.4	Experiment 13: R_COMP FREE confirmation	35
6.4.1	Provisional results (ERROR_CERT vs MAPPED, non-confirmatory)	35
6.5	Experiment 14: Freeze and GG screening	35
6.5.1	V0 frozen: development baselines	35
6.5.2	GG checker	36
6.6	Evaluation 4: Corrected record	36
6.7	Iteration 5 Summary	36
6.8	Final dead ends	37
7	What We Have Learned	37
8	Related Work	39

1 Framing

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning pipelines [10], yet evaluating whether a candidate formula faithfully captures a sentence’s meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: Brunello et al. [3] found that approximately 42% of entries in both FOLIO [1] and MALLS [2] contain incorrect formalisations. Existing metrics either require a gold formula (exact match, prover-checked equivalence) or only verify that the output parses [5, 6]. The user’s own pilot study used structural consistency checks (arity agreement, predicate-set overlap across reruns, joint satisfiability) that measure stability but not whether a formula says what its sentence says.

This run investigates gold-free metrics that predict faithfulness to the text given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge. The evaluation is a meta-evaluation: we build a labelled set of (sentence, formula) pairs from public benchmarks whose gold has been audited, and measure each metric’s ability to discriminate correct translations from errors.

2 Iteration 1

2.1 Strategy

The iteration screens four candidate metric families head-to-head on a frozen evaluation set, with a disguised cheap LLM judge as the bar to beat.

Candidates. (A) FOL-Triage, a three-layer cascade: L1 (formula lint: deterministic formula smells), L2 (text accounting: bag-of-words and role-aware coverage), and L3 (role questionnaire: a small LLM reads the text and z3 reads the formula; their structured answers are compared). (B) Round-trip verification: verbalise the formula back to natural language and compare to the original via NLI or embedding similarity [7]. (C) Cross-system consensus: collect translations of the same sentence from multiple independent systems, measure z3-equivalence between the candidate and the peer pool, and score disagreement [8]. (D) Disguised cheap judge and baselines: an LLM judge, structural pilot metrics, and round-trip features evaluated on nonce-disguised sentences to control for memorisation.

Evaluation set. The frozen screen uses Logic-LM [10] released outputs on FOLIO validation: three LLM systems (GPT-3.5-turbo, GPT-4, text-davinci-003) producing 796 items. Track L contains LLM outputs labelled by z3 equivalence to corrected gold [4]; track H contains the 294 original human gold formulas, of which 57 are errors according to the corrected labels. The parallel run measured 43.9% of solver-non-equivalent candidates as faithful by a three-family panel; iteration 2 re-scores with those adjudicated labels.

Decision rule. A candidate must achieve $\text{AUROC} \geq 0.65$ on track L (ERROR vs CORRECT), with coverage $\geq 90\%$, rewrite false-alarm rate $\leq 10\%$ per rewrite family, and cost $\leq \$0.002$ per item. It advances only if its AUROC exceeds the disguised cheap judge’s, with a paired bootstrap CI excluding zero.

Development dataset (Artifact E). A separate 700-sentence, 8,507-row development set was constructed from FOLIO training premises, MALLS training long/conditional/exception strata, and FOLIO curated conclusions, each translated by 9 LLM families plus ccg2lambda plus GPT-4 gold. Labels were assigned by z3 equivalence to audited gold, with contested cases adjudicated by a three-member panel (Claude Haiku-4.5, GLM-4.6, Kimi-K2). This set is reserved for iteration 2 confirmation.¹

¹Artifact `gen_art_dataset_1.`

Code: <https://github.com/ai-inventor-papers/>

The dataset contains 1,750 CORRECT, 5,005 ERROR, 1,086 UNPARSEABLE, 276 UNRESOLVED, and 390 CONTESTED items. The panel calibration subset (173 items) shows that the panel agrees with expert-corrected labels on both faithful and unfaithful items. A screen audit of 1,173 items from the Logic-LM screen confirms the label assignment. The dataset cost \$9.83 in API calls.

2.2 Experiment A: FOL-Triage

FOL-Triage was evaluated on the frozen screen with Gemini 2.5 Flash Lite as the primary LLM for L3. Total API spend was \$0.29.²

2.2.1 Headline AUROCs (track L, ERROR vs CORRECT, $n = 477$, 199 errors)

Table 1: FOL-Triage headline AUROCs on track L (Iteration 1, Experiment A).

Metric	AUROC	95% CI	AUPRC
L1 (any smell)	0.508	[0.499, 0.519]	0.425
L2 bag-of-words	0.737	[0.677, 0.801]	0.690
L2 role-aware	0.557	[0.527, 0.594]	0.476
L3 score	0.756	[0.701, 0.809]	0.657
Fused (fitted on H, no leak)	0.759	[0.695, 0.825]	0.740
Fused (all H)	0.827	[0.778, 0.874]	0.771
Fused (cross-fit L)	0.864	[0.824, 0.902]	0.789
Decomposed judge (LLM reads formula)	0.716	[0.645, 0.789]	0.717
L3 with local Qwen2.5-1.5B	0.708	[0.637, 0.774]	0.603

The primary no-leak fused score is 0.759. The fused-all-H and cross-fit-L numbers (0.827, 0.864) are shown for completeness but are not valid held-out estimates: the former uses the full track H for fitting, and the latter is a within-L cross-fit that overstates generalisation.

Track H (human gold errors, $n = 258$, 57 errors). Fused out-of-fold AUROC is 0.777 [0.698, 0.847]. L1 alone reaches 0.655, compared to 0.508 on track L. The lint rules fire on human annotation errors but not on LLM outputs, because the two populations produce different formula smells.

2.2.2 L1 lint: a negative result on LLM outputs

L1 flags formulas with deterministic smells (existential implication, free variables, arity clashes, vacuity, trivial formulas, dangling predicates). On track L, the true positive rate is 2.0% at a false positive rate of 0.4%, giving AUROC 0.508. This is a negative result: L1 lint does not transfer from human annotation errors (where it is effective, AUROC 0.655) to LLM outputs. The reason is that LLM-generated formulas rarely contain the syntactic pathologies L1 detects; their errors are semantic (wrong predicates, missing conditions, scope mistakes).

All nine individual smell rules have false-alarm rates below 3% on every system:

ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-1/
dataset-1

²Artifact gen.art.experiment.1. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-1/experiment-1

Table 2: L1 lint per-smell false-alarm rates by system (Iteration 1, Experiment A).

Smell	gpt-3.5 FA	gpt-4 FA	davinci-003 FA
EX_IMP	0.0	0.0	0.0
ALL_AND	0.0	0.0	0.0
IFF_RESTR	0.0	0.009	0.0
GLUE	0.0	0.0	0.0
FREE	0.0	0.0	0.0
VACUOUS	0.0	0.0	0.0
TRIVIAL	0.0	0.0	0.0
ARITY	0.0	0.0	0.0
DANGLING	0.0	0.0	0.0

Only the FREE smell fires on any track-L errors (4 items).

2.2.3 L2 role-aware check: below bag-of-words

The role-aware UD-based check (L2-role) achieves AUROC 0.557, which is 0.18 below the simpler bag-of-words check (L2-bow, 0.737). The role-aware layer adds no discriminative power over word overlap.

2.2.4 Decomposed judge ablation

Replacing z3 with an LLM that reads the formula directly (the decomposed judge) drops L3 AUROC from 0.756 to 0.697 ($\Delta\text{AUROC} = -0.059$). The LLM gets only 57.6% of binary argument slots correct on track L and 36.4% on track H. The z3 answerer is significantly more accurate for structured formula reading.

2.2.5 Per-type sensitivity (track L, at binary threshold)

Table 3: FOL-Triage per-type sensitivity on track L (Iteration 1, Experiment A).

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	5	0.0	0.2	1.0	0.2	0.4	1.0
QUANT	2	0.0	0.0	0.5	0.0	0.5	0.5
CONN	4	0.0	0.0	0.0	0.0	0.0	0.0
DROP	142	0.028	0.479	0.127	0.197	0.599	0.606
ADD	150	0.020	0.473	0.107	0.173	0.653	0.573
SWAP	12	0.0	0.0	0.083	0.25	0.083	0.333
BIND	8	0.125	0.0	0.125	0.0	0.375	0.25
COMPOUND	187	0.118	0.428	0.166	0.160	0.599	0.578

FOL-Triage is most sensitive to ADD and DROP errors (the largest categories), with the fused score catching about 60%. It misses CONN errors entirely (0/4) and detects SWAP poorly (1/12). Negation errors are fully caught by the cascade (which fires L2-role, where monotonicity shifts are visible) but L1 and L2-bow miss them.

2.2.6 Per-type sensitivity (track H)

Table 4: FOL-Triage per-type sensitivity on track H (Iteration 1, Experiment A).

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	4	0.0	0.0	0.25	0.0	0.25	0.25
REV	1	0.0	0.0	0.0	0.0	0.0	0.0
QUANT	1	0.0	0.0	0.0	0.0	0.0	0.0
RESTR	1	1.0	1.0	0.0	0.0	1.0	1.0
CONN	8	0.25	0.125	0.0	0.375	0.375	0.375
MOVE	2	0.5	0.0	0.5	0.0	0.5	1.0
DROP	14	0.143	0.357	0.0	0.143	0.357	0.5
ADD	19	0.158	0.368	0.053	0.211	0.526	0.579
SWAP	2	0.0	0.5	0.0	0.5	0.5	0.5
BIND	1	1.0	1.0	0.0	0.0	1.0	1.0
UNGLUE	5	1.0	0.4	0.0	0.2	1.0	1.0
COMPOUND	33	0.303	0.182	0.242	0.182	0.545	0.636

On track H (57 errors), the per-type pattern differs. UNGLUE errors ($n = 5$) are caught at 100% by L1 and the fused score, because they produce detectable formula smells.

2.2.7 Invariance under meaning-preserving rewrites

Five rewrite families were tested on 150 track-L CORRECT items.

Table 5: Rewrite false-alarm and flip rates for FOL-Triage layers (Iteration 1, Experiment A).

Rewrite	n	L1 FA	L1 flip	bow FA	bow flip	role FA	role flip	L3 FA	L3 flip	Fused FA	Fused flip
RENAME	150	0.0	0.0	0.18	0.093	0.013	0.0	0.033	0.007	0.22	0.013
REORDER	34	0.0	0.0	0.029	0.0	0.059	0.0	0.059	0.0	0.235	0.0
DEMORGAN	33	0.0	0.0	0.030	0.0	0.061	0.0	0.030	0.030	0.182	0.061
CONTRA	40	0.025	0.025	0.10	0.0	0.025	0.025	0.025	0.0	0.275	0.0
PRENEX	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

The rewrite false-alarm gate ($\leq 10\%$ per family) fails. The worst-family fused FA is 0.275 (CONTRAPOSITIVE), but this equals the FA on the original formulas; the flip rate is at most 0.061. The high FA is not caused by the rewrites but by the fused score’s calibration: it was fitted on track H where the CORRECT base rate is higher, producing a systematic over-flagging on track L (FA on track-L CORRECT items is 0.194 versus the 0.10 calibration target).

L1, L2-role, and L3 individually pass the 10% gate. L2-bow fails only under RENAME (FA = 0.18), because renaming predicates breaks the bag-of-words match between text and formula.

2.2.8 Coverage

Table 6: FOL-Triage coverage rates (Iteration 1, Experiment A).

Track	n	Parseable	Full coverage rate
L	753	693	0.920
H	302	279	0.924

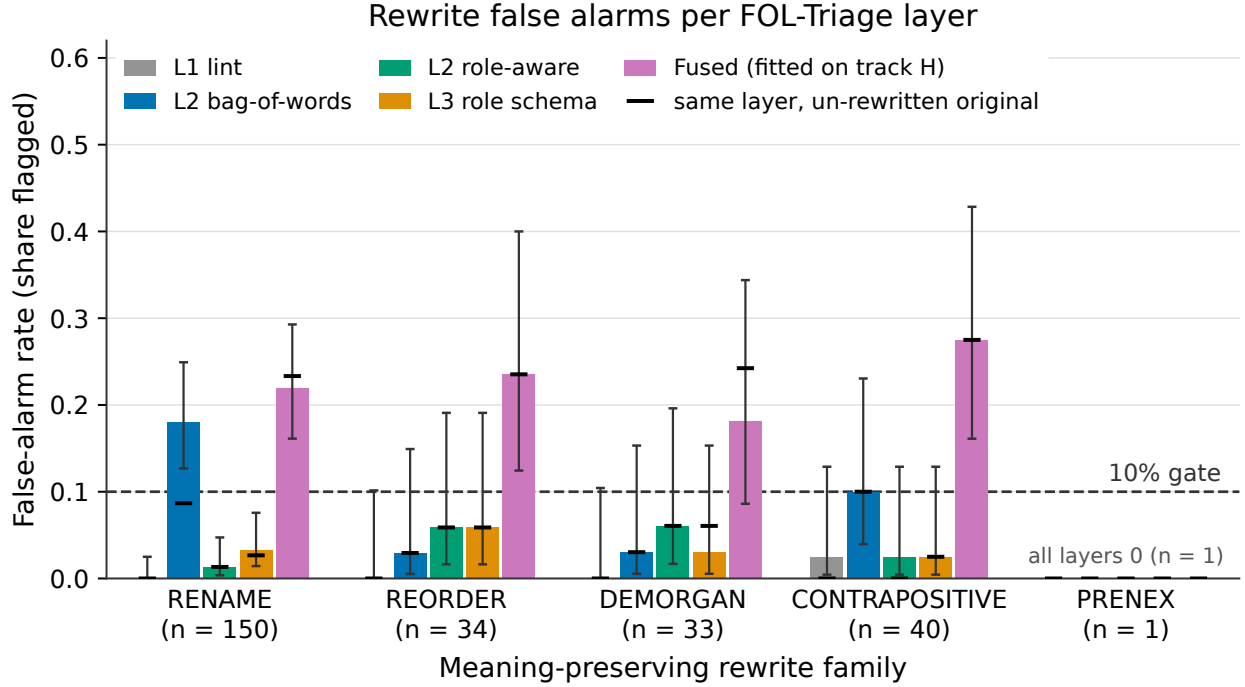


Figure 1: False-alarm (FA) rate of each FOL-Triage layer under five families of meaning-preserving rewrites, measured on the 150 track-L CORRECT items. Bars give the share of rewritten formulas flagged by L1 lint (grey), L2 bag-of-words (blue), L2 role-aware (green), L3 role schema (amber) and the fused score fitted on track H (violet); whiskers are Wilson 95% intervals, and black ticks mark the same layer’s FA on the un-rewritten originals. The dashed line is the pre-registered 10% gate. The fused score fails the gate in every family with $n \geq 33$ (0.18–0.28), but it flags the un-rewritten originals at about the same rate, and its flip rate is at most 0.061.

Parse rates by system: gpt-3.5-turbo 0.889, gpt-4 0.946, text-davinci-003 0.921. The coverage gate ($\geq 90\%$) passes.

2.2.9 Cost

L1 and L2 are free (CPU-only). L3 costs \$0.175 per 1,000 calls with Gemini 2.5 Flash Lite. The total per-item cost is \$0.0002, passing the \$0.002 gate.

2.2.10 Contamination check

L3 was run on nonce-disguised sentences on the full track L. AUROC with exact text-formula alignment: 0.765; with disguised text: 0.770. Paired difference: -0.005 $[-0.063, 0.049]$. There is no evidence that the LLM benefits from having seen the public benchmark sentences.

2.2.11 Complexity strata (track L, fused score)

Table 7: Complexity strata AUROCs on track L (Iteration 1, Experiment A).

Stratum	n (error/correct)	Fused AUROC	L2-bow	L3
words < 12	443 (179/264)	0.781	0.742	0.772
words 12–19	31 (18/13)	0.521	0.596	0.541
words ≥ 20	3 (2/1)	0.500	1.000	0.000
0 quantifiers	302 (130/172)	0.743	0.728	0.771
1 quantifier	164 (65/99)	0.794	0.748	0.721
≥ 2 quantifiers	11 (4/7)	0.750	0.768	0.607
depth ≤ 1	296 (124/172)	0.762	0.745	0.779
depth 2–3	173 (73/100)	0.756	0.717	0.710
depth ≥ 4	8 (2/6)	0.667	0.750	0.500
0 conditions	344 (145/199)	0.775	0.759	0.791
1–2 conditions	131 (54/77)	0.716	0.677	0.628

The frozen screen is dominated by short, simple sentences (93% under 12 words). The long and complex strata have too few items to test. This limitation motivates the development dataset (Artifact E), which oversamples these strata.

2.2.12 Per-system AUROC (descriptive)

Table 8: Per-system AUROCs on track L (Iteration 1, Experiment A).

System	n	Error rate	Fused	L1	L2-bow	L3
gpt-3.5-turbo	143	0.378	0.689	0.500	0.650	0.722
gpt-4	177	0.367	0.700	0.496	0.716	0.720
text-davinci-003	157	0.510	0.857	0.525	0.811	0.817

FOL-Triage is most discriminative on text-davinci-003, the weakest system.

2.3 Experiment C: Cross-System Consensus

The consensus metric collects translations of the same sentence from multiple independent LLM families, checks pairwise z3 equivalence, and scores each candidate by its disagreement with the pool. Total API spend was \$1.52.³

2.3.1 Headline AUROCs (track L, ERROR vs CORRECT, $n = 546$, 249 errors)

Table 9: Consensus metric AUROCs on track L (Iteration 1, Experiment C).

Metric	AUROC	95% CI	AUPRC
c_score (1 - eq_frac, all peers)	0.866	[0.821, 0.906]	0.827
c_score (6 fresh peers only)	0.872	[0.830, 0.911]	0.825
c_score (non-OpenAI peers)	0.863	[0.820, 0.903]	0.808
c_score (2 Logic-LM peers only)	0.753	[0.701, 0.806]	0.659
medoid depth	0.722	[0.673, 0.774]	0.622
cluster entropy	0.773	[0.724, 0.820]	0.691
SC-5 cheap (gpt-4.1-nano)	0.725	[0.669, 0.781]	0.627
SC-5 same model	0.716	[0.632, 0.796]	0.628
SC-5 entropy	0.631	[0.568, 0.693]	0.539
Pred-set instability (SC)	0.688	[0.626, 0.750]	0.602
Pred-set instability (pool)	0.770	[0.717, 0.821]	0.721
FOL length	0.527	[0.470, 0.586]	0.485
Misalignment (vocab)	0.835	[0.794, 0.876]	0.821

The `c_score` achieves the highest AUROC in the iteration at 0.866. It beats SC-5 (the self-consistency baseline) by $\Delta\text{AUROC} = +0.140$ [0.085, 0.197]. Using only the six fresh peer translations gives 0.872, slightly higher than using all peers. Restricting to the two other Logic-LM systems drops the AUROC to 0.753, showing that diversity across model families is essential.

2.3.2 RENAME rewrite invariance: failure

Under predicate renaming (replacing predicate names with WordNet synonyms or nonce names), the `c_score` false-alarm rate is 96% (100% for constant-only renaming). This is expected and fundamental: two formulas with different predicate names are not z3-equivalent, so any rename breaks the equivalence check.

2.3.3 Blind-spot analysis

38% of ERROR items are z3-equivalent to the medoid (the most popular peer formula). This places a ceiling on recall of approximately 62%. The blind-spot rate is 5% for polarity errors and 57% for structural errors.

2.3.4 Cost

The peer pool costs \$0.0015 per sentence (6 peers via API). Total experiment spend: \$1.52. z3 pairwise equivalence checking is free.

³Artifact `gen_art_experiment_3`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqkX/round-1/experiment-3

Consensus metric AUROCs, iteration-1 screen (track L, $n = 546$)

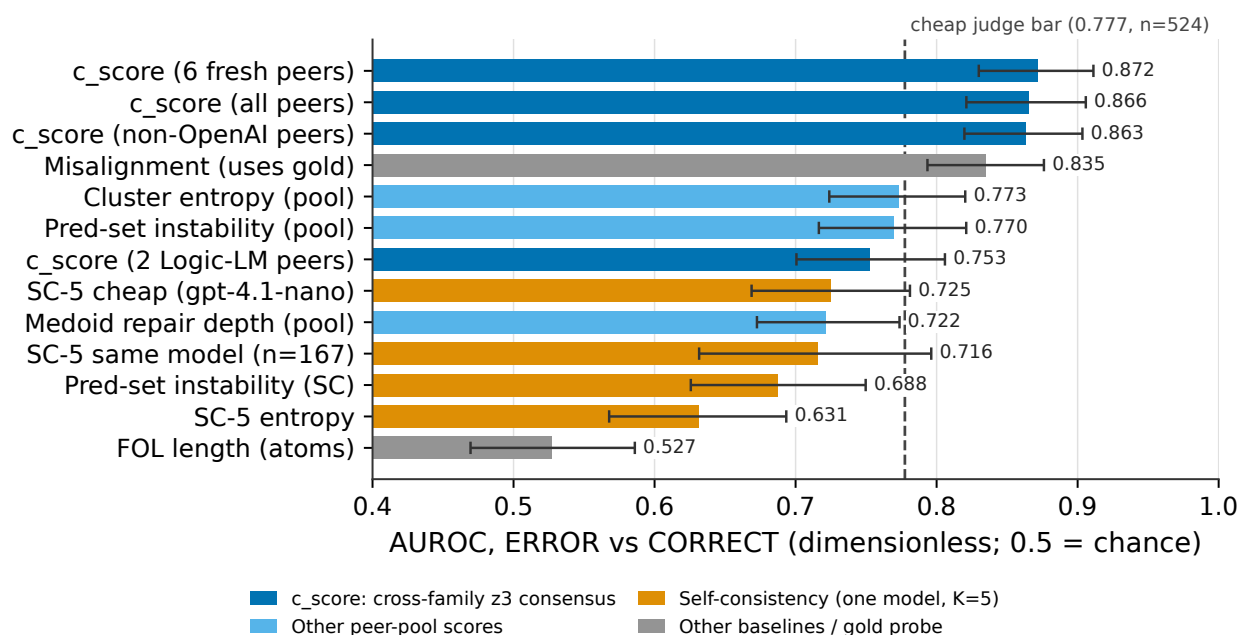


Figure 2: AUROC (ERROR vs. CORRECT) of consensus-based scores and baselines on the iteration-1 frozen screen (track L, $n = 546$, 249 errors). Bars are sorted by AUROC. Dark blue: c_score variants. Light blue: other pool scores. Orange: self-consistency scores. Grey: FOL length and misalignment probe. The dashed line marks the disguised cheap LLM judge (AUROC 0.777).

2.4 Experiment D: Disguised Judge and Baselines

This experiment establishes the bar: every baseline metric and every judge variant on the same shared screen, scored with nonce-disguised sentences to control for memorisation. Total API spend was \$2.02.⁴

The primary cheap judge (Gemini 2.5 Flash Lite, JSON 0–100 scoring, nonce-disguised sentences) achieves AUROC 0.777 [0.720, 0.827] on track L ($n = 524$, 230 errors, 294 correct).

Table 10: Full baseline table on track L (Iteration 1, Experiment D).

Metric	AUROC	95% CI
parse_fail	0.500	[0.500, 0.500]
pilot_joint_conflict	0.513	[0.488, 0.539]
pilot_arity_incons	0.500	[0.500, 0.500]
pilot_shape_incons	0.524	[0.470, 0.578]
pilot_dangling	0.533	[0.474, 0.592]
pilot_undeclared	0.525	[0.510, 0.542]
pilot_rerun_jacc	0.720	[0.659, 0.780]
rt_nli_min	0.710	[0.648, 0.769]
rt_nli_fwd	0.652	[0.588, 0.713]
rt_nli_bwd	0.681	[0.621, 0.742]
rt_nli_contra	0.628	[0.566, 0.686]
rt_nli_min_alt	0.689	[0.627, 0.749]
rt_embed_cos	0.633	[0.574, 0.696]
rt_reformalise_eq	0.513	[0.479, 0.547]
judge_cheap_orig	0.749	[0.695, 0.801]
judge_cheap_disg	0.777	[0.720, 0.827]
judge_cheap2_orig (gpt-4.1-nano)	0.750	[0.691, 0.810]
judge_cheap2_disg	0.714	[0.648, 0.777]
judge_strong_orig (gemini-3.1-pro)	0.802	[0.715, 0.877]
judge_strong_disg	0.730	[0.655, 0.803]
judge_local_qwen8b_disg	0.757	[0.702, 0.811]
judge_local_llama8b_disg	0.671	[0.604, 0.737]
judge_local_qwen14b_disg	0.760	[0.681, 0.834]
rt_nli_min_localverb	0.740	[0.680, 0.796]

2.4.1 Pilot structural metrics: negative result

The user’s pilot structural metrics are null on this screen. Joint conflict achieves AUROC 0.513, arity inconsistency 0.500, shape inconsistency 0.524, dangling predicates 0.533, and undeclared predicates 0.525.

2.4.2 Round-trip verification

Round-trip NLI achieves AUROC 0.710 [0.648, 0.769], 0.067 below the cheap judge ($p = 0.008$ by DeLong test). The round-trip reformalisation check achieves only 0.513, because 76% of items tie at the same score.

⁴Artifact `gen.art.experiment_4` Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqkX/round-1/experiment-4

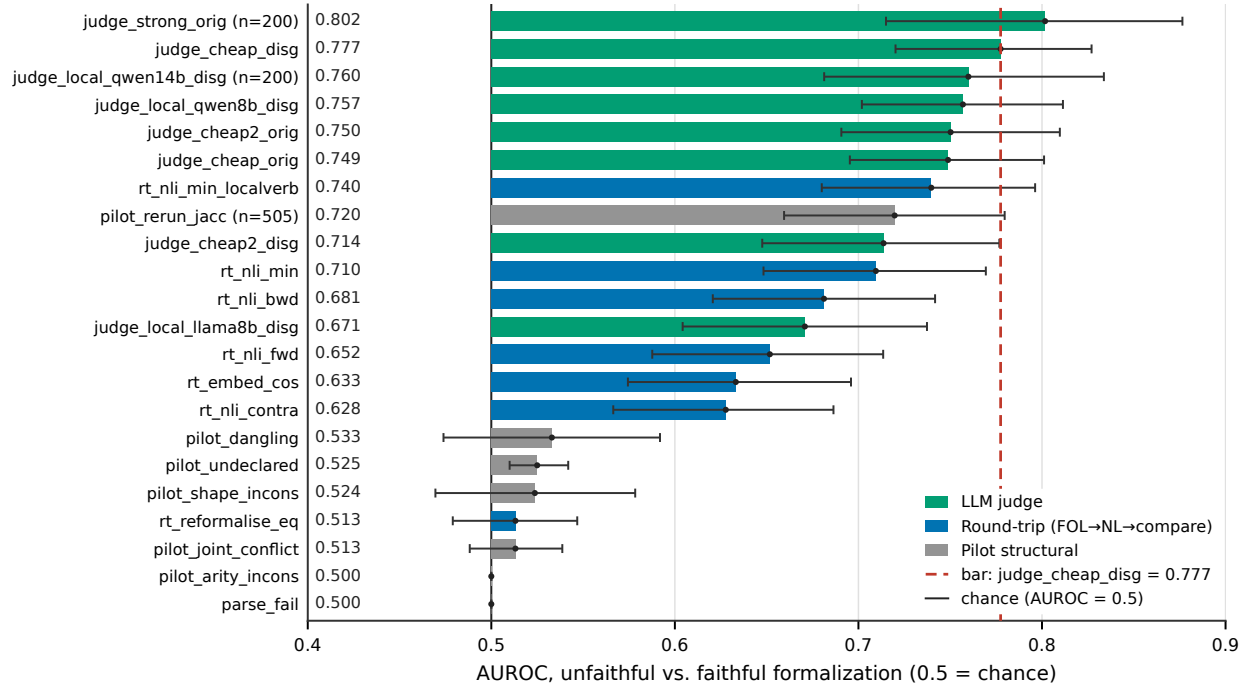


Figure 3: AUROC for separating unfaithful (ERROR) from faithful (CORRECT) candidate formalizations on the iteration-1 screen (track L, $n = 524$, 230 errors). Bars coloured by family: LLM judges (green), round-trip metrics (blue), pilot structural metrics (grey). The disguised flash-lite judge sets the bar at 0.777 (red dashed line).

2.4.3 Cross-fitted baseline combinations (track L)

Table 11: Cross-fitted baseline combinations on track L (Iteration 1, Experiment D).

Combination	AUROC	95% CI
S1 (structural)	0.698	[0.634, 0.761]
S2 (round-trip)	0.687	[0.625, 0.749]
S3 (judges disguised)	0.771	[0.705, 0.831]
S4 (all cheap features)	0.817	[0.762, 0.867]
S6 (S4 + local judges)	0.810	[0.753, 0.861]

S4 (all cheap features combined via cross-fitted logistic regression) achieves 0.817, +0.039 $[-0.010, 0.087]$ over the cheap judge alone (not significant), setting the empirical ceiling for cheap baselines.

2.4.4 Contamination analysis

No evidence of contamination was found for any judge. All DiD CIs include zero:

Table 12: Contamination DiD test for judges (Iteration 1, Experiment D).

Judge	DiD (H – L)	95% CI
flash-lite	−0.042	[−0.170, 0.094]
gpt-4.1-nano	−0.077	[−0.188, 0.023]
gemini-3.1-pro	+0.002	[−0.115, 0.104]
Qwen3-8B	−0.045	[−0.161, 0.079]
Llama-3.1-8B	+0.023	[−0.112, 0.157]
Qwen3-14B	−0.073	[−0.195, 0.056]

2.4.5 Invariance of judges under meaning-preserving rewrites

Judges are highly sensitive to meaning-preserving rewrites:

Table 13: Judge flip rates under meaning-preserving rewrites (Iteration 1, Experiment D).

Metric	RENAME	REORDER	DEMORGAN	CONTRA	ALL
judge_cheap_orig	0.27	0.12	0.23	0.64	0.29
judge_cheap_disg	0.44	0.12	0.46	0.41	0.39
judge_cheap2_orig	0.49	0.12	0.46	0.95	0.49
rt_nli_min	0.37	0.10	0.31	0.33	0.31
rt_embed_cos	0.57	0.07	0.44	0.38	0.44
FOL-Triage fused	0.013	0.0	0.061	0.0	0.013

FOL-Triage has the lowest flip rate of any metric tested. Judges and round-trip methods change their verdict on 29–49% of meaning-preserving rewrites.

Table 14: Per-error-type sensitivity of judges on track H ($n = 46$ errors, Iteration 1, Experiment D).

Group	n	cheap_orig	cheap_disg	cheap2_orig	rt_nli_min
1-op errors	13	0.308	0.538	0.462	0.231
2-op errors	11	0.545	0.818	0.727	0.273
ADD	12	0.500	0.750	0.667	0.250
COMPOUND	22	0.545	0.864	0.773	0.273
DROP	11	0.455	0.636	0.455	0.182
CONN	3	0.667	0.667	1.000	0.333
NEG	2	0.500	0.500	1.000	0.500
SWAP	2	0.000	1.000	0.500	0.000

2.4.6 Per-error-type sensitivity of judges (track H)

2.4.7 Complexity analysis (track L)

Table 15: Judge complexity strata on track L (Iteration 1, Experiment D).

Stratum	n (err/cor)	judge_cheap_disg	rt_nli_min
words < 12	485 (210/275)	0.784	0.713
words 12–19	36 (18/18)	0.667	0.670
quantifiers 0–1	512 (226/286)	0.784	0.712
depth ≤ 3	466 (199/267)	0.781	0.702
depth 4–5	54 (30/24)	0.762	0.828
0 conditions	394 (182/212)	0.783	0.745
1–2 conditions	128 (48/80)	0.765	0.653

2.5 Iteration 1 Summary

Table 16: Iteration 1 summary: four candidates and the bar.

Method	AUROC	95% CI	n	Cost/item	Flip rate
c_score (consensus)	0.866	[0.821, 0.906]	546	\$0.0015/sent	96% (RENAME)
S4 baseline combo	0.817	[0.762, 0.867]	524	varies	varies
Bar: judge_cheap_disg	0.777	[0.720, 0.827]	524	\$0.00004	0.39
FOL-Triage fused	0.759	[0.695, 0.825]	477	\$0.0002	0.013
rt_nli_min	0.710	[0.648, 0.769]	524	\$0.00001	0.31
L1 lint	0.508	[0.499, 0.519]	477	\$0	0.0
Pilot structural	0.50–0.53	n/a	524	\$0	n/a

What iteration 1 established:

1. The cross-system consensus metric ($\text{c_score} = 1 - \text{eq_frac}$) achieves the highest AUROC at 0.866, beating SC-5 by +0.140. It requires peer translations from multiple model families and is fundamentally non-invariant to predicate renaming (96% FA under RENAME).

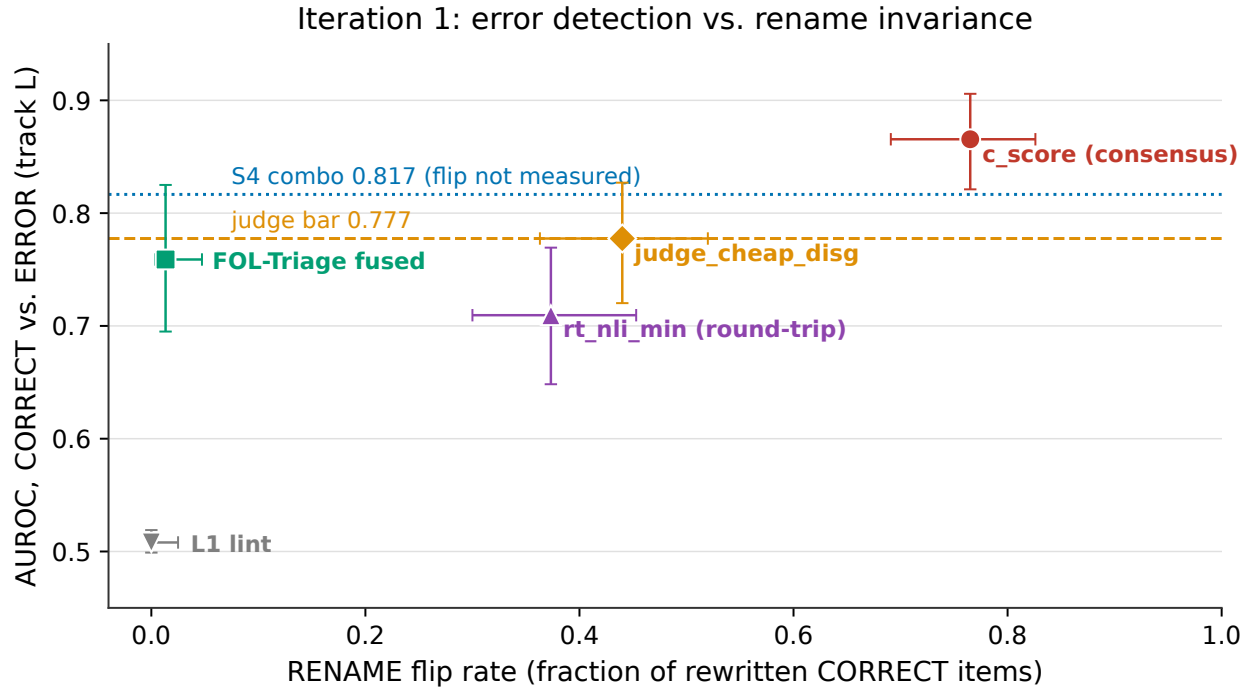


Figure 4: Discrimination versus rename invariance for the iteration-1 metric families. The y-axis is AUROC on track L; the x-axis is the RENAME flip rate. Points: c_score (red circle, AUROC 0.866), the disguised cheap LLM judge (orange diamond, 0.777), FOL-Triage fused (green square, 0.759), round-trip rt_nli_min (purple triangle, 0.710) and L1 lint (grey triangle, 0.508).

2. The disguised cheap judge sets the bar at AUROC 0.777. The S4 combination reaches 0.817 (+0.039, not significant). The frontier judge adds nothing significant under solver labels; under panel labels the advantage strengthens (+0.166, $p < 0.001$).
3. FOL-Triage achieves AUROC 0.759, below the bar. Its primary value is invariance: the fused flip rate under rewrites is 0.013, compared to 0.29–0.49 for judges. However, the fused false-alarm rate on CORRECT items is 0.194.
4. L1 lint does not transfer from human annotation errors to LLM outputs. The pilot structural metrics are null. Round-trip NLI is 0.067 below the judge.
5. The evaluation labels are noisy: 58.8% of track-L “errors” are pure ADD+DROP (likely vocabulary mismatches).
6. No contamination was detected for any LLM-based metric.

3 Iteration 2

3.1 Strategy

Iteration 2 has three goals: (i) confirm iteration 1’s ordering on the development dataset E under both the solver labels (R_AB, R_A) and panel-adjudicated labels (R_ADJ); (ii) test whether fusing the consensus signal (PEER) with text-based checks (TEXT = L2-bow + L3) improves over each alone; and (iii) construct a typed perturbation suite (PERTURB) for controlled sensitivity testing.

The pre-registered bar for this iteration is the flash-lite judge on disguised sentences, with the local Qwen3-8B judge as fallback. The shared OpenRouter key hit its daily limit before the development-set experiment completed; the flash-lite bar is therefore untestable on E. The fallback local judge (Qwen3-8B, rubric B, greedy; screen AUROC 0.757 vs 0.777 for flash-lite) is the labelled secondary bar.

3.2 Experiment 5: PEER+TEXT on development dataset E

PEER+TEXT is a logistic fusion of the consensus signal (`c_score_align`) and the text-based FOL-Triage layers (L2-bow, L3 questionnaire). The fusion weights were fitted on the iteration-1 screen and frozen before any E score was computed. API spend was \$0.⁵

3.2.1 Name-free alignment: failed the pre-registered screen gate

A name-free variant (graded clause consensus, `g_score`) was pre-registered as the primary PEER signal. Both name-free variants (NF-pure and NF-anchored) failed the screen gate: for both, more than 10% of AGREE track-L CORRECT items tie at $g = 1$. Post-hoc fractional tie-breaking showed this failure was a threshold-tie artefact. NF-anchored passes both gates under fractional scoring (RENAME FA 0.074, ROLE_PERMUTE recall 0.771); NF-pure still fails. The NF-anchored fusion on E (R_AB pooled) achieves AUROC 0.759 [0.72, 0.79], 0.031 below the pre-registered PEER+TEXT fusion (0.790).

Table 17: PEER+TEXT and component AUROCs on development set E (Iteration 2, Experiment 5).

Subset	n (err/cor)	PEER+TEXT	c_score_align	NF c_score	L2-bow	L3	Local judge
R_AB pooled	2686 (1822/864)	0.790	0.782	0.743	0.734	0.579	0.712
R_AB L25	873 (697/176)	0.755	0.712	0.651	0.722	0.560	0.664
R_AB L20	821 (592/229)	0.747	0.722	0.643	0.715	0.547	0.683
R_AB EXC	606 (384/222)	0.777	0.814	0.782	0.669	0.620	0.663
R_AB CTRL	386 (149/237)	0.708	0.742	0.741	0.613	0.490	0.739
R_AB long pool	2300 (1673/627)	0.768	0.759	0.704	0.717	0.580	0.653
R_A L20+EXC	449 (261/188)	0.818	0.860	0.802	0.678	0.587	0.554

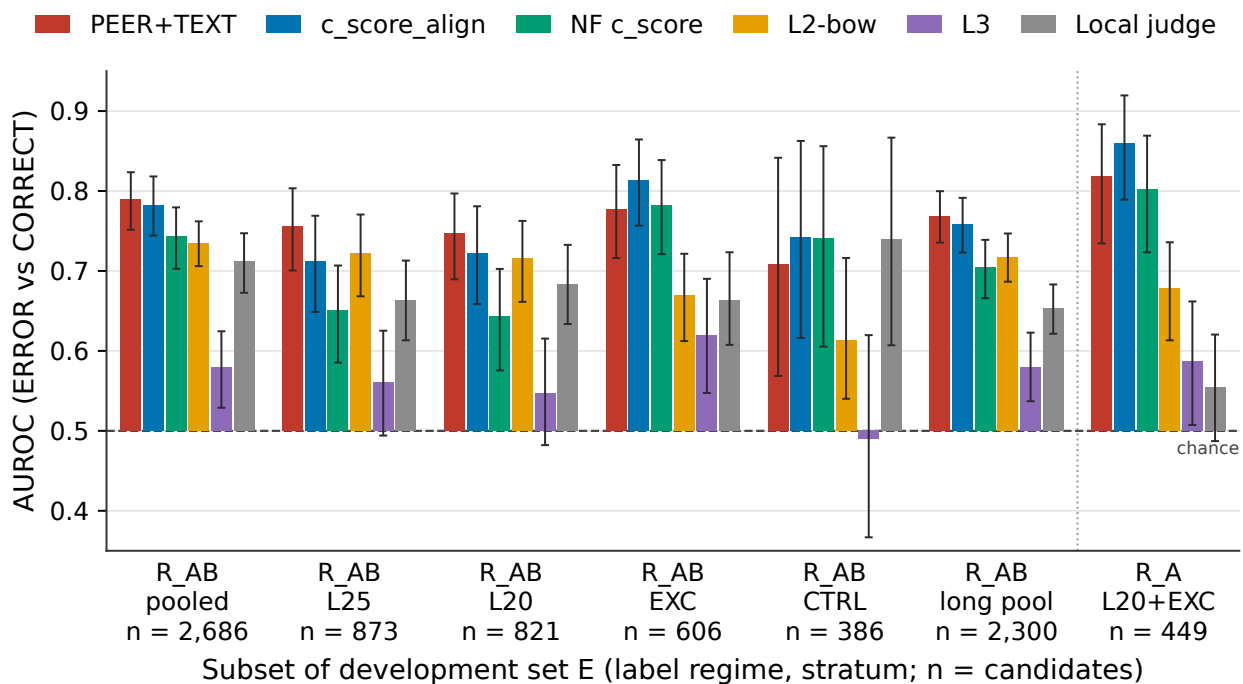


Figure 5: Item-level AUROC of six gold-free metrics on development dataset E. Bars show PEER+TEXT (red), c_score_align (blue), NF c_score (green), L2-bow (orange), L3 (purple) and the local Qwen3-8B judge (grey). PEER+TEXT scores highest on the pooled set (0.790). On every long-sentence subset both consensus metrics beat the local judge.

3.2.2 Headline AUROCs on E (R_AB, ERROR vs CORRECT)

The paired Δ AUROC of PEER+TEXT minus the local judge is +0.078 [0.043, 0.112] on R_AB pooled (DeLong $p < 0.0001$). On the long pool, the gap widens to +0.115 [0.075, 0.157].

3.2.3 Does fusion beat c_score_align alone?

The stratified AUROC shows PEER+TEXT at 0.753 and c_score_align at 0.741, a difference of +0.011 [−0.012, 0.038], not significant. The TEXT layers add no significant discriminative power over c_score_align alone once the stratum-composition confound is removed.

3.2.4 Mechanism predictions

Four pre-registered mechanism predictions were tested on E:

P1 (PEER catches COVERAGE errors that TEXT misses, and vice versa): INCONCLUSIVE.

P2 (PEER and TEXT error rankings are uncorrelated): REFUTED. Spearman correlation between PEER and TEXT among errors is 0.294.

P3 (PEER+TEXT advantage grows with sentence complexity): INCONCLUSIVE.

P4 (PEER+TEXT is more invariant than judges): REFUTED. The fused score’s RENAME FA is 0.385 vs the local judge’s 0.683. Better than judges but far worse than FOL-Triage fused (FA 0.22).

3.2.5 Rewrite false alarms (screen, CORRECT bases)

Table 18: Rewrite false-alarm rates on screen CORRECT bases (Iteration 2, Experiment 5).

Rewrite	n	Fused	c_score_align	NF g	L2-bow	L3	Local judge
RENAME	104	0.385	0.859	0.074	0.216	0.168	0.683
REORDER	27	0.222	0.178	0.037	0.044	0.444	0.000
DEMORGAN	26	0.154	0.185	0.038	0.046	0.231	0.231
CONTRA	28	0.107	0.063	0.237	0.075	0.188	0.214

3.2.6 System-level correlation

Kendall τ -b of each metric vs the R_AB system error rate across 13 system $\times$ variant rows: PEER+TEXT 0.718, TEXT alone 0.769, PEER alone 0.692, local judge 0.667.

3.3 Evaluation 1: Re-evaluation under alternative label regimes

All 57 reviewer-audited quantities from the iteration-1 report reproduce exactly under independent recomputation (57/57 MATCH).⁶

⁵Artifact `gen.art.experiment_5`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-2/experiment-5

⁶Artifact `gen.art.evaluation_1`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-2/evaluation-1

Table 19: Regime AUROC matrix on the common set (Iteration 2, Evaluation 1).

Metric	R_SOLVER	R_ADJ_AB	R_ADJ_A	R_ADJ_ALL
PT (PEER+TEXT)	0.871	0.877	0.932	0.906
fused_H (FOL-Triage)	0.770	0.868	0.958	0.866
c_score	0.849	0.779	0.786	0.831
L2-bow	0.732	0.815	0.917	0.810
L3	0.761	0.718	0.813	0.751
judge_cheap_disg	0.782	0.772	0.751	0.816
judge_cheap_orig	0.754	0.797	0.858	0.785
sc5_cheap	0.713	0.703	0.676	0.737
rt_nli_min	0.709	0.749	0.731	0.760

Under R_ADJ_AB, PEER+TEXT (0.877) leads, followed by FOL-Triage fused (0.868) and L2-bow (0.815). The `c_score` drops from 0.849 (solver) to 0.779 (R_ADJ_AB), because the adjudicated labels reclassify many vocabulary-mismatch items as CORRECT.

3.3.1 Label regime shifts

Table 20: Label regime shift on 161 shared items (Iteration 2, Evaluation 1).

Metric	Shift	95% CI	Direction
fused_H	+0.080	[0.003, 0.171]	UP
L2-bow	+0.102	[0.018, 0.192]	UP
c_score	-0.106	[-0.255, 0.013]	INSIDE CI
judge_cheap_disg	+0.018	[-0.094, 0.134]	INSIDE CI

3.4 Dataset 3: R_COMP and PERTURB

This artifact constructs two evaluation resources: R_COMP (long composed sentences with trusted FOL references) and PERTURB (a typed perturbation suite with z3-verified controls).⁷

R_COMP: incomplete (OpenRouter key exhausted). The R_COMP component generates long, complex sentences (≥ 25 words, ≥ 3 conditions) by composing atoms from a verified lexicon of 652 entries. Nine templates (T1–T9) produce sentences with weak and strong exception readings, each with a z3-verified reference formula. The candidate generation, labelling, and adjudication phases did not run because the shared OpenRouter key hit its \$50 daily limit.

PERTURB: complete.

3.5 Experiment 6: Baselines on development set E

Experiment 6 scored all local and baseline metrics on development dataset E under R_AB labels ($n = 2686$, 1822 ERROR / 864 CORRECT, 292 sentences).

⁷Artifact `gen_art_dataset_3`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05Bqxx/round-2/dataset-3

Table 21: PERTURB suite statistics (Iteration 2, Dataset 3).

Statistic	Value
Base formulas	300
Total mutants (DOWN + UP)	4,234
Matched controls	868
Total suite rows	5,102
Complete DOWN/UP pairs	1,354
z3 verification failures	0
Operator types	12
Cost	\$0.41

Table 22: Pooled AUROCs on E (R_AB, selected metrics; Iteration 2, Experiment 6).

Metric	AUROC [95% CI]
S4_local (all tiers)	0.754 [0.715, 0.789]
S4_local (R_AB)	0.748 [0.709, 0.782]
judge_local_qwen8b_disg	0.710 [0.674, 0.746]
S4_local_noLLMjudge	0.703 [0.660, 0.742]
sc5_local_eq_frac	0.664 [0.617, 0.706]
pilot_rerun_jacc	0.660 [0.621, 0.695]
judge_local_qwen8b_orig	0.649 [0.604, 0.693]
judge_local_llama8b_disg	0.636 [0.590, 0.681]
rt_nli_min_local	0.626 [0.580, 0.673]
b2_score	0.574 [0.529, 0.619]
pilot_shape_incons	0.504 [0.499, 0.510]
parse_fail	0.500 [0.500, 0.500]

3.6 Dead ends: B2 and R_ADJ

Candidate B2 (decomposed z3 distinguishing-worlds reader): DROPPED. The gate balanced accuracy was 0.476 (target ≥ 0.85). The b2_score AUROC on E (R_AB pooled) was 0.574, below every other non-null metric.

R_ADJ adjudicator gate: DROPPED.

Table 23: R_ADJ adjudicator gate results (Iteration 2).

Adjudicator	Gate balanced accuracy	Target
Sonnet-5	0.662	≥ 0.80
Grok-4.20	0.527	≥ 0.80

Inter-adjudicator kappa was 0.497 on all items and 0.416 on real E rows. Sonnet-5 test-retest kappa was 0.834 with a flip rate of 0.07.

3.7 Iteration 2 Summary

What iteration 2 established:

1. PEER+TEXT achieves AUROC 0.790 [0.75, 0.82] on development dataset E (R_AB pooled, $n = 2686$), beating the local judge by +0.078 [0.043, 0.112].
2. Fusion does not significantly beat `c_score_align` alone. The stratified Δ AUROC is +0.011 [-0.012, 0.038].
3. The name-free consensus variant failed the pre-registered screen gate due to a threshold-tie artefact.
4. Under adjudicated labels, FOL-Triage fused rises from 0.770 (solver) to 0.868 (adjudicated). The `c_score` drops from 0.849 to 0.779.
5. The frontier judge advantage is large under adjudicated labels (+0.166 AUROC, $p < 0.001$).
6. All 57 reviewer-audited numbers reproduce exactly.
7. The PERTURB suite (4,234 typed mutants + 868 controls) is complete.

Iteration-2 total API spend: \$4.229.

4 Iteration 3

4.1 Strategy

Iteration 3 resolves two gaps from iteration 2: the flash-lite API bar and the R_COMP candidate generation. It runs five pre-registered tests: T1 (head-on API bar), T2 (R_COMP), T3 (PERTURB per-operator sensitivity), T4 (mechanism verification), and T5 (PERTURB rename tradeoff).

4.2 Test T1: Head-on API bar on development set E

The shared OpenRouter key was available. All API metrics were scored on E.⁸

4.2.1 Verdict

Table 24: T1 verdict summary (Iteration 3).

Criterion	Status	Key numbers
(a) c_score_align beats flash-lite disg	CONFIRMED	strat Δ +0.099 [+0.049, +0.146]
(b) adds signal beyond S4_full	CONFIRMED	strat Δ +0.039 [+0.021, +0.057]
(d) frontier ratio ≥ 0.95	CONFIRMED	ratio 0.957; cost 30.2 $\times$ cheaper
VEX confound control	CONFIRMED	strat Δ +0.306 [+0.232, +0.376]
M3 length slope	DISCONFIRMED	words Δ slope +0.146 [−0.051, +0.361]

Overall T1: CONFIRMED. c_score_align (strat AUROC 0.741) exceeds the flash-lite disguised judge (0.642) by +0.099 [+0.049, +0.146] on development set E.

4.2.2 Item-level AUROC on R_AB (E_POOL PRIMARY)

Table 25: Item-level AUROC on E, R_AB (Iteration 3, Experiment 6).

Metric	Pooled [CI]		Strat [CI]	AUPRC	\$/item
p_peer_text	0.790	[0.752, 0.825]	0.753	[0.720, 0.784]	0.879 1.21e-04
c_score_align	0.782	[0.745, 0.816]	0.741	[0.705, 0.775]	0.861 1.21e-04
S4_full_oof	0.777	[0.739, 0.809]	0.735	[0.699, 0.766]	0.874 –
S4_local_oof	0.748	[0.709, 0.782]	0.693	[0.656, 0.729]	0.856 –
nf_c_score	0.743	[0.702, 0.780]	0.686	[0.647, 0.720]	0.829 –
L2-bow	0.734	[0.705, 0.762]	0.697	[0.667, 0.726]	0.830 –
judge_local_qwen8b_disg	0.710	[0.674, 0.746]	0.674	[0.646, 0.703]	0.815 –
judge_cheap2_orig	0.700	[0.659, 0.739]	0.653	[0.616, 0.688]	0.823 2.01e-05
judge_cheap_disg	0.696	[0.655, 0.733]	0.642	[0.607, 0.676]	0.786 4.88e-05
judge_cheap2_disg	0.681	[0.641, 0.719]	0.644	[0.609, 0.680]	0.806 2.01e-05
sc5_eq_frac	0.656	[0.609, 0.699]	0.598	[0.559, 0.637]	0.758 2.95e-05
rt_nli_min	0.639	[0.593, 0.687]	0.606	[0.569, 0.645]	0.776 2.75e-05
judge_cheap_orig	0.635	[0.595, 0.671]	0.623	[0.593, 0.651]	0.772 4.88e-05
L3	0.579	[0.532, 0.625]	0.562	[0.524, 0.601]	0.720 –
rt_embed_cos	0.574	[0.527, 0.623]	0.566	[0.526, 0.607]	0.731 –
parse_fail	0.500	[0.500, 0.500]	0.500	[0.500, 0.500]	0.678 –

4.2.3 Paired deltas: challenger minus bar (stratified)

On CTRL sentences (short, simple), the consensus advantage vanishes. The advantage is largest on exception sentences (EXC: +0.197) and on R_A tier-A labels (+0.299).

⁸Artifact `gen_art_experiment_6`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-3/experiment-6

Table 26: Paired deltas vs flash-lite disguised, stratified (Iteration 3).

Cell	n (E/C)	Challenger	Δ vs flash-lite disg [CI]
R_AB pooled	1822/864	c_score_align	+0.099 [+0.049, +0.146]
R_AB pooled	1822/864	p_peer_text	+0.110 [+0.065, +0.154]
R_AB long	1673/627	c_score_align	+0.116 [+0.070, +0.163]
R_AB long	1673/627	p_peer_text	+0.132 [+0.091, +0.173]
R_AB EXC	384/222	c_score_align	+0.197 [+0.124, +0.269]
R_A L20+EXC	261/188	c_score_align	+0.299 [+0.212, +0.390]
R_AB CTRL	149/237	c_score_align	-0.069 [-0.232, +0.098]

4.2.4 Cross-fitted stacks and nesting

Table 27: Cross-fitted stacks on E (Iteration 3).

Stack	Pooled OOF [CI]	Strat OOF [CI]
S4_local	0.748 [0.709, 0.782]	0.693 [0.656, 0.729]
S4_full	0.777 [0.739, 0.809]	0.735 [0.699, 0.766]
S4_full + c_score_align	0.807 [0.772, 0.838]	0.774 [0.743, 0.802]
S4_full + p_peer_text	0.808 [0.775, 0.839]	0.776 [0.745, 0.805]
S4_local + c_score_align	0.796 [0.759, 0.828]	0.757 [0.723, 0.788]
PT_refit (4 features)	0.808 [0.769, 0.842]	0.778 [0.742, 0.813]

Adding `c_score_align` to `S4_full` yields +0.039 strat AUROC (permutation null $p_{95} = 0.009$). The 4-feature `PT_refit` model matches the full `S4` stack without any judge features.

4.2.5 VEX confound control

On VEX items (183 ERROR / 433 CORRECT, 146 sentences): strat Δ `c_score_align` minus `judge_cheap_disg` = +0.306 [+0.232, +0.376]. Even when all names match, consensus outperforms the judge by a wide margin.

4.2.6 AUROC by complexity bin

Table 28: AUROC by word-count bin on E (Iteration 3).

Bin	n (E/C)	c_score_align	p_peer_text	judge_disg	S4_full	Δ c-judge [CI]
<12	154/283	0.725	0.746	0.766	0.772	-0.040 [-0.178, +0.103]
12-19	255/134	0.701	0.682	0.607	0.667	+0.094 [+0.013, +0.188]
20-24	686/262	0.727	0.740	0.625	0.715	+0.103 [+0.026, +0.171]
25-34	614/161	0.705	0.753	0.618	0.727	+0.087 [-0.006, +0.178]
≥ 35	113/24	0.759	0.756	0.706	0.789	+0.054 [-0.115, +0.173]

The consensus advantage is concentrated in sentences with 12+ words, and is largest on exception-clause sentences (AUROC 0.964 vs 0.600 for “except” sentences).

Table 29: Recall per error group at matched FA 0.10 (Iteration 3).

Group	n	c_align	p_peer_text	judge_disg	judge_strong	S4_full
COMPOUND (tier B)	1148	0.55	0.63	0.25	0.67	0.56
polarity	434	0.53	0.56	0.26	0.74	0.53
structural	1282	0.49	0.53	0.24	0.60	0.50
COVERAGE	246	0.27	0.26	0.13	0.27	0.20
MEANING_RENAME	221	0.07	0.10	0.17	0.09	0.17
PEER_ENDORSED	398	0.00	0.06	0.16	0.07	0.17

4.2.7 Recall per error group at matched FA 0.10

Consensus has the highest recall on COMPOUND and polarity errors but near-zero recall on MEANING_RENAME and PEER_ENDORSED errors.

4.2.8 System-level correlation

Table 30: System-level Kendall τ -b (Iteration 3).

Metric	τ -b system $\times$ variant	τ -b families
p_peer_text	0.718	0.822
c_score_align	0.692	0.822
judge_cheap2_disg	0.718	0.822
judge_cheap_disg	0.564	0.600
S4_full_oof	0.590	0.644

4.3 Test T2: R_COMP (long composed sentences)

R_COMP generates long, complex sentences (≥ 25 words, 4–5 conditions) from a verified lexicon, each with a z3-verified reference formula.⁹ 221 sentences produced 2,024 candidate translations. Population: 429 ERROR / 1,595 CORRECT.

4.3.1 SIG: primary test

Table 31: SIG primary test result (Iteration 3, Experiment 7).

Statistic	c_score_sig	judge_disg	Δ [CI]	PASS
Within-template AUROC	0.954	0.587	+0.367 [+0.328, +0.404]	True

The consensus metric’s within-template AUROC is 0.954, massively outperforming the flash-lite disguised judge at 0.587. The delta of +0.367 is the largest effect observed in any test.

⁹Artifact `gen_art_experiment_7`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-3/experiment-7

4.3.2 SIG: per-metric AUROCs

Table 32: Per-metric AUROCs on R_COMP SIG (Iteration 3, Experiment 7).

Metric	n	Within-template [CI]	Pooled [CI]	AUPRC
g_align	2024	0.977 [0.970, 0.983]	0.978 [0.972, 0.983]	0.878
g_nf	2024	0.976 [0.968, 0.983]	0.976 [0.970, 0.982]	0.877
c_score_sig	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
c_score_align	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
judge.frontier.orig	60	0.932 [0.782, 1.000]	0.932 [0.855, 0.990]	0.881
judge.frontier.disg	60	0.729 [0.573, 0.867]	0.735 [0.593, 0.851]	0.655
judge.cheap.orig	1932	0.673 [0.637, 0.708]	0.710 [0.678, 0.741]	0.345
judge.cheap.disg	1904	0.587 [0.551, 0.622]	0.578 [0.537, 0.617]	0.238
judge.local.disg	2024	0.498 [0.459, 0.534]	0.541 [0.505, 0.575]	0.233

4.3.3 Rename invariance on R_COMP

Table 33: Rename invariance on R_COMP (Iteration 3, Experiment 7).

Metric	Unrenamed FA	RENAME_NONCE FA	RENAME_SYN FA
c_score_sig	0.260	1.000	1.000
c_score_align	0.260	1.000	0.635
c_score_nf	0.263	0.261	0.252
c_score_hyb	0.260	0.261	0.252

Under SIG, nonce-renaming causes c_score_sig and c_score_align to flag every item (FA = 1.0). The NF and HYB variants are invariant. This confirms the fundamental trade-off.

4.3.4 FREE condition: NOT_TESTABLE

The FREE condition produced only 34 CORRECT tier-A items against 420 ERROR, with 1,811 UNRESOLVED. FREE is NOT_TESTABLE and deferred to iteration 4.

4.4 Tests T3 + T5: PERTURB sensitivity and rename tradeoff

The PERTURB suite was scored with all metrics.¹⁰

4.4.1 T3: Per-operator within-base AUROC

c_align achieves within-base AUROC ≥ 0.84 on every testable operator. MEANING_RENAME is an ERROR operator; c_nf is blind to it (0.499, chance).

Table 34: Per-operator within-base AUROC on PERTURB (Iteration 3, Experiment 8).

Operator	n	c_align	c_hyb	c_nf	rt_nli_min	judge_disg
NEG	379	0.869	0.884	0.843	0.873	0.723
REV	179	0.860	0.869	0.826	0.578	0.697
QUANT	179	0.860	0.872	0.829	0.351	0.632
RESTR	177	0.859	0.873	0.827	0.608	0.638
CONN	327	0.861	0.870	0.832	0.563	0.663
DROP	191	0.841	0.841	0.791	0.792	0.728
ADD	472	0.859	0.860	0.820	0.895	0.683
SWAP	116	0.882	0.910	0.863	0.454	0.529
BIND	314	0.865	0.858	0.798	0.833	0.672
MEANING_RENAME	318	0.872	0.685	0.499	0.930	0.600

Table 35: Rename tradeoff on PERTURB E.bases (Iteration 3, Experiment 8).

Metric	NONCE FA	SYN FA	MR wb-AUROC	SWAP recall	BASE FA
c_align	0.765	0.565	0.872	0.758	0.201
c_nf	0.132	0.201	0.499	0.512	0.163
c_hyb	0.216	0.298	0.685	0.832	0.197
g_align	0.413	0.318	0.841	0.426	0.154
g_nf	0.132	0.176	0.477	0.428	0.154
rt_nli_min	0.686	0.191	0.930	0.153	0.143
judge_cheap_disg	0.229	0.132	0.600	0.220	0.129

4.4.2 T5: Rename tradeoff table

The rename tradeoff is stark. c_align has RENAME_NONCE FA of 0.765: 76.5% of correctly translated formulas are flagged as errors after predicate renaming. c_nf has 0.132 (invariant) but sacrifices SWAP recall (0.512 vs 0.758). No consensus variant achieves both low rename FA and high structural-error recall.

4.5 Test T4: Mechanism verification

48 numerical claims from the iteration-2 report were independently recomputed. All 48 match their source values (48/48 VERIFIED_MATCH).¹¹

Mechanism predictions: M1 INCONCLUSIVE, M2 CONFIRMED but non-specific, M3 INCONCLUSIVE, M4 **CONFIRMED** ($k_{95} = 3$; by words tercile $k_{95} = 2/3/5$).

At $k = 3$, the AUROC is 0.757, already 96.6% of the full-pool value (0.784 at $k = 7$).

¹⁰Artifact `gen_art_experiment_8`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-3/experiment-8

¹¹Artifact `gen_art_evaluation_2`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-3/evaluation-2

Table 36: k-curve on development set E (Iteration 3, Evaluation 2).

k	AUROC	AUROC words T3	\$/sentence
1	0.685	0.581	\$0.000315
3	0.757	0.651	\$0.000946
5	0.776	0.698	\$0.001576
7	0.784	0.726	\$0.002206

4.6 Prior-art positioning

A systematic prior-art search was conducted to position the cross-family consensus metric against the literature.¹² The search covered 63 sources with 139 verbatim quotes machine-checked against fetched text. No result tables were produced; the artifact’s output is a structured verdict on five claims.

Scoop rule. The pre-stated rule requires all three of: (i) multiple model families’ formal translations, (ii) solver equivalence, and (iii) meta-evaluation of the agreement score against faithfulness labels on NL→FOL. No prior paper meets all three. The method is NOT scooped as a method-plus-evaluation package, but the method itself (cross-model agreement) is not new.

Closest neighbours. (1) ARc [19] translates NL into SMT-LIB with several LLMs and scores each translation by the proportion of k translations that entail it — the same functional form as `c_score`. It uses a fixed shared schema and validates on downstream QA, not faithfulness labels. (2) NoTB [20] clusters RTL from four LLM families by formal equivalence; precision rises from 63% to 94.7% at ≥ 4 agreeing families (coverage 27%), but reports no AUROC and no complexity analysis. (3) GenV [21] trains a single NL→FOL verifier achieving AUROC 0.961 on z3-reference labels; its single-model self-consistency baseline reaches 0.863; critically, its LLM judge wins (0.778 vs 0.679) when labels switch from z3-reference to panel intent. (4) SCP-NL2TL [22] reports error-detection AUROC of single-model self-consistency vs judge by difficulty tier for NL→temporal logic.

Table 37: Prior-art claim verdicts (Iteration 3, Artifact `gen_art_research_1`).

Claim	Verdict	Key qualifier
C1: beats API judge	NEEDS-QUALIFIER	Method not new; claim first meta-evaluation
C2: holds with aligner-free labels	SAFE	No neighbour controls for shared instrument
C3: rename-invariant matcher	NEEDS-QUALIFIER	Alignment exists; rename FA rate is novel
C4: e/d decomposition	NEEDS-QUALIFIER	Scatter premise known since 1978
C5: 3–5 families suffice	NEEDS-QUALIFIER	Already recommended; per-tercile is new

4.7 Iteration 3 Summary

What iteration 3 established:

1. **T1 CONFIRMED:** `c_score_align` (strat AUROC 0.741) exceeds the flash-lite disguised judge (0.642) by +0.099 [+0.049, +0.146].

¹²Artifact `gen_art_research_1`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqkX/round-3/research-1

2. **T1 CONFIRMED:** adding `c_score_align` to `S4_full` yields +0.039 [+0.021, +0.057] strat AUROC.
3. **T2 NOT CONFIRMED:** SIG within-template AUROC 0.954 vs judge 0.587 ($\Delta = +0.367$), but SIG is the out-of-scope condition. FREE is NOT_TESTABLE.
4. **T3:** Consensus recall at the PRIMARY threshold is flat across operators.
5. **T5:** `c_align` RENAME_NONCE FA 0.765; `c_nf` 0.132 but sacrifices SWAP recall.
6. **T4:** M4 CONFIRMED ($k_{95} = 3$); M3 INCONCLUSIVE. 48/48 numerical claims verified.
7. Dead ends confirmed: B2 DROPPED, R_ADJ DROPPED, FREE NOT_TESTABLE.

Iteration-3 API spend: \$4.33.

5 Iteration 4

5.1 Strategy

Iteration 4 tests Candidate-Signature Consensus (CSC): instead of having peers translate independently, CSC feeds each peer the candidate formula’s own predicate symbols and asks it to translate using those symbols. If the candidate is correct, the peers should converge; if wrong, they may be anchored into reproducing its error. CSC is tested on development set E (T6-E) and on PERTURB (T6-P). In parallel, the E2 confirmation dataset is constructed (550 sentences) and name-free labels are generated for R_COMP FREE candidates.

5.2 Experiment 9: CSC on development set E — provisional negative result

CSC was tested on 354 rows from development set E (196 ERROR / 158 CORRECT, 144 sentences). The budget event at \$0.108 stopped generation early.¹³

5.2.1 Anchoring: the core finding

Table 38: CSC vs FREE anchoring comparison (Iteration 4, Experiment 9).

Component	CSC	FREE_exact (matched)
e (error endorsement)	0.459	0.173
d (correct-row divergence)	0.139	0.323

CSC dramatically lowers d (correct translations converge when given the candidate’s vocabulary) but raises e from 0.173 to 0.459 (nearly half of all errors are endorsed by anchored peers). The net effect is negative.

5.2.2 Headline AUROCs on E

CSC (0.614) is below every established metric including the flash-lite judge (0.664). Gates G1 and G2 fail. CSC adds no signal to `S4_full` (+0.006 [−0.013, +0.025]).

¹³Artifact `gen.art.experiment-9`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-4/experiment-9

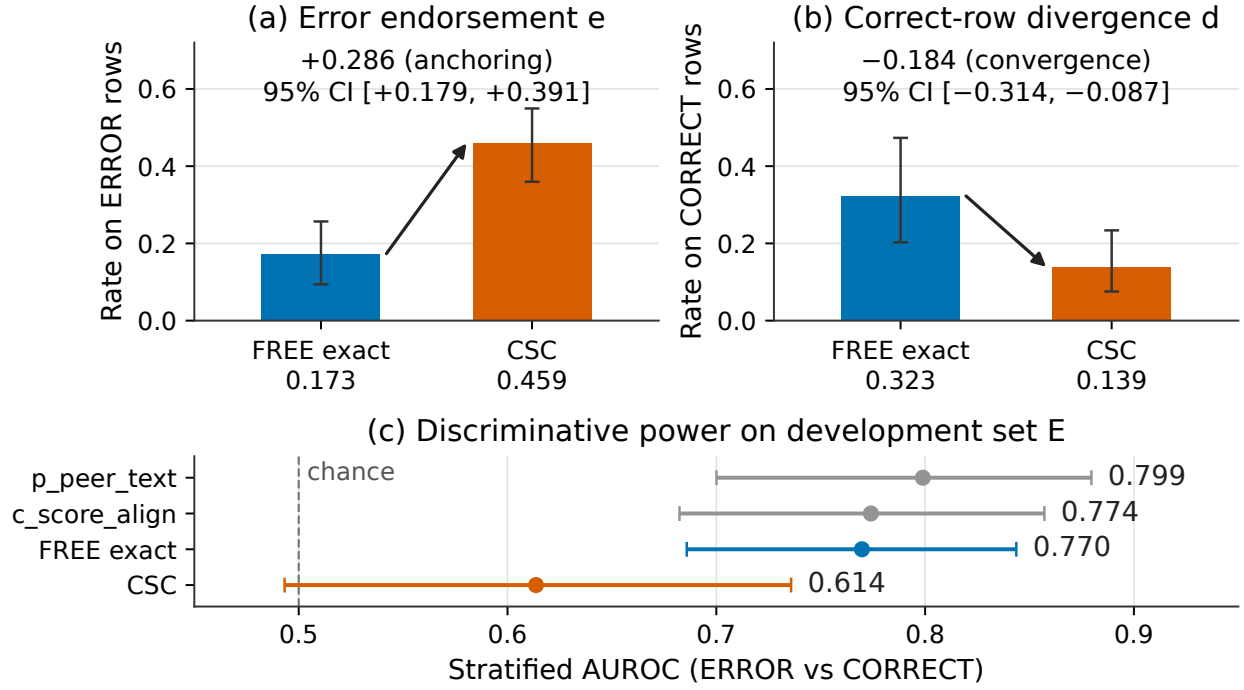


Figure 6: Effect of Candidate-Signature Consensus (CSC, vermillion) versus FREE exact consensus (blue) on the PRIMARY subset of development set E (354 rows: 196 ERROR, 158 CORRECT). (a) Error endorsement e : 0.173 $\rightarrow$ 0.459, a paired change of +0.286 (+0.179, +0.391). (b) Correct-row divergence d : 0.323 $\rightarrow$ 0.139, a paired change of -0.184 (-0.314, -0.087). (c) Stratified AUROC: CSC 0.614 vs FREE 0.770.

Table 39: Headline stratified AUROCs on E PRIMARY (Iteration 4, Experiment 9).

Metric	Strat AUROC [CI]
p_peer_text	0.799 [0.700, 0.880]
c_score_align	0.774 [0.682, 0.857]
FREE_exact (matched)	0.770 [0.686, 0.844]
S4_full	0.736 [0.635, 0.826]
flash-lite judge	0.664 [0.568, 0.750]
CSC	0.614 [0.493, 0.736]

5.2.3 Anchoring by error class

Table 40: Anchoring by error class (Iteration 4, Experiment 9).

Error class	e_{CSC}	$e_{\text{FREE_exact}}$
MEANING_RENAME	0.821	0.036
ADD	0.490	0.135
COMPOUND	0.282	0.107

5.3 Experiment 10: CSC on PERTURB — budget exhausted

The run-wide OpenRouter budget (\$7.00) was exhausted before experiment 10’s first CSC peer call. The artifact spent \$0.0000088 (one 1-token probe).¹⁴

5.3.1 SIGPROXY: cued consensus on R_COMP

Table 41: SIGPROXY cued consensus on R_COMP (Iteration 4, Experiment 10).

Metric	n	Within-template AUROC [CI]
c_proxy (K3)	2024	0.930 [0.916, 0.943]
c_proxy_graded	2024	0.938 [0.926, 0.950]
c_score_sig (full)	2024	0.952 [0.936, 0.966]
g_align	2024	0.977 [0.970, 0.983]
judge_cheap_disg	1904	0.587 [0.550, 0.624]
judge_local_disg	2024	0.498 [0.462, 0.534]

The 3-family cued consensus (0.930) reaches 98% of the full SIG consensus (0.952), consistent with M4.

5.3.2 Same-vocabulary typed repair

Table 42: Same-vocabulary typed repair accuracy (Iteration 4, Experiment 10).

Target	Base source	n	Strict accuracy [CI]
oracle (known vocab)	ALL	4234	0.921 [0.915, 0.928]
proxy (SIGPROXY)	ALL	1196	0.784 [0.711, 0.849]
free3 (independent vocab)	E only	3884	0.123 [0.081, 0.169]

Error typing requires shared vocabulary; under independent vocabulary, accuracy drops to 12.3%.

¹⁴Artifact `gen.art.experiment_10`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqkX/round-4/experiment-10

5.4 Dataset 4: E2 confirmation set — partial

E2 is a fresh 550-sentence confirmation set constructed from MALLS-train and ProverQA-dev.¹⁵

Status: PARTIAL (budget-stopped). The run-wide OpenRouter budget was exhausted. E2 completed a 30-sentence pilot (300 candidate rows); the remaining 520 sentences are pending. All confirmation cells have zero CORRECT rows. Projected cost to complete: \$6.79.

5.5 Dataset 5: R_COMP FREE name-free labels

This artifact produces name-free labels for the 2,652 FREE candidate translations on R_COMP using exhaustive map-family search.¹⁶

ERROR_CERT: 759. MAPPED: 1,506. UNPARSEABLE: 188. NO_OUTPUT: 199. The CORRECT class is empty (gloss step not run), so FREE AUROC is NOT_TESTABLE.

5.5.1 Executor audit

Table 43: Executor audit of FREE labels (Iteration 4, Dataset 5).

Class	Result
ERROR_CERT	51/60 UNFAITHFUL, precision 0.867 [0.758, 0.931]
MAPPED	52/60 faithful = 0.867 [0.758, 0.931]

ERROR_CERT precision (0.867) is below the 0.90 flag.

5.6 Evaluation 3: T8 vocabulary analysis

The T8 evaluation decomposes the consensus mechanism by comparing cross-family agreement under shared vocabulary (SIG) vs independent vocabulary (FREE) on R_COMP.¹⁷

Table 44: Cross-family agreement floor (Iteration 4, Evaluation 3).

Statistic	Value
END_MAJ d	0.415
Predicted d_{floor}	0.402 [0.336, 0.471]
R_exact AUROC	0.683
No-anchoring oracle floor	0.385
e ceiling	0.162

SIG-minus-FREE cross-family exact-agreement drop: 0.479 [0.447, 0.513]. Sharing a vocabulary raises pairwise exact agreement from ~5% (FREE) to ~53% (SIG).

¹⁵Artifact `gen_art_dataset_4.` Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxxX/round-4/dataset-4

¹⁶Artifact `gen_art_dataset_5.` Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxxX/round-4/dataset-5

¹⁷Artifact `gen_art_evaluation_3.` Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxxX/round-4/evaluation-3

5.7 Iteration 4 Summary

What iteration 4 established:

1. **CSC is a provisional negative.** It lowers d (0.139 vs 0.323) but raises e (0.459 vs 0.173). Strat AUROC 0.614 is below all established metrics. The mechanism is anchoring: sharing the candidate’s vocabulary anchors peers into reproducing its errors.
2. Cued consensus confirms k-saturation: SIGPROXY 0.930 (98% of full SIG 0.952).
3. Vocabulary sharing reduces d by 0.61–0.84 on R_COMP while maintaining perfect mutant recall.
4. Error typing requires shared vocabulary: 0.784–0.921 accuracy (shared) vs 0.085 (independent).
5. E2 is constructed but not testable: 550 sentences frozen, 520 pending generation.
6. R_COMP FREE labels: search-only, gloss pending. FREE AUROC is NOT_TESTABLE.

5.8 Dead ends accumulated

Table 45: Dead ends accumulated through iteration 4.

Dead end	Iter	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557 (below bag-of-words)
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC (provisional)	4	Strat AUROC 0.614, anchoring $e = 0.459$

6 Iteration 5

6.1 Strategy

Iteration 5 is the final iteration. It attempts four things: (1) run the E2-A confirmation on the untouched 110-sentence subset from dataset 4; (2) extend E2 to long sentences only (E2-B, 400 sentences ≥ 25 words); (3) complete the R_COMP FREE confirmation by running the gloss gate; and (4) freeze V0 as the shipped variant and screen five consensus variants (V1–V5) against it on development set E, alongside a Gloss-Gated (GG) name-free agreement checker.

6.2 Experiment 11: E2-A confirmation

E2-A is the untouched portion of the E2 confirmation set: 110 sentences (37 L25, 37 EXC, 36 DT) with labels joined after both seal hashes matched.¹⁸

¹⁸Artifact `gen.art.experiment-11`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-5/experiment-11

Overall verdict: NOT_TESTABLE. The pre-registered flash-lite bar has zero scored rows in R_AB.

6.2.1 DT stratum (testable)

The DT cell (179 rows: 74 ERROR / 105 CORRECT, 26 sentences) is testable. V0 strat AUROC 0.916 [0.858, 0.970].

Table 46: E2-A DT stratum AUROCs (Iteration 5, Experiment 11).

Metric	DT strat AUROC [CI]
c_score_align (V0)	0.916 [0.858, 0.970]
p_peer_text	0.919 [0.860, 0.976]
S4_E2	0.953 [0.911, 0.986]
judge_local_qwen8b_disg	0.913 [0.858, 0.958]
l3_z3	0.572 [0.480, 0.675]

V0 minus judge_local on DT: $\Delta = +0.003$ $[-0.042, +0.052]$ (n.s.).

6.2.2 Long pool (L25 + EXC, not testable)

The long pool has 94 rows (76 ERROR / 18 CORRECT, 12 sentences): not testable. V0 strat AUROC 0.603 [0.463, 0.735]. L25 alone (42 rows, 7 CORRECT): V0 strat AUROC 0.443 [0.375, 0.500], near chance.

6.2.3 ALL cell

On ALL 273 rows (150 ERROR / 123 CORRECT, 38 sentences): V0 strat AUROC 0.890 [0.838, 0.940].

6.3 Experiment 12: E2-B (long-sentence pool)

E2-B extends E2 with 400 sentences ≥ 25 words from MALLS-train.¹⁹

Status: NOT_TESTABLE. Labels are solver-vs-unaudited-MALLS-gold only (R_A_UNAUDITED). Label counts: 48 CORRECT, 334 ERROR.

Table 47: E2-B exploratory AUROCs under R_A_UNAUDITED (Iteration 5, Experiment 12).

Metric	Strat AUROC [CI]
V0 (c_score_align)	0.737 [0.645, 0.827]
V2 (reliability-weighted)	0.750 [0.659, 0.838]
V4 (two-channel logistic)	0.764 [0.666, 0.851]
p_peer_text	0.655 [0.546, 0.767]
flash-lite disguised	0.496 [0.356, 0.620]
S4_E2B	0.498 [0.402, 0.594]

¹⁹Artifact `gen_art_experiment_12`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqkX/round-5/experiment-12

V0 minus flash-lite: $\Delta = +0.208$ [$+0.066$, $+0.363$].

6.4 Experiment 13: R_COMP FREE confirmation

This experiment attempts the in-scope FREE condition of R_COMP.²⁰

Gloss gate: FAILED twice. Run 1: haiku BA 0.776, qwen 0.855, both_yes 0.817. Run 2: haiku BA 0.561, qwen 0.883, both_yes 0.765. All below 0.90. Every MAPPED row remains UNRESOLVED.

Confirmatory verdict: NOT_TESTABLE. Zero CORRECT rows exist.

6.4.1 Provisional results (ERROR_CERT vs MAPPED, non-confirmatory)

Table 48: R_COMP FREE provisional AUROCs (Iteration 5, Experiment 13).

Metric	Within-template AUROC [CI]
c_score_align	0.801 [0.770, 0.834]
c_score_hyb	0.817 [0.788, 0.848]
c_score_nf	0.743 [0.710, 0.776]
c_exact	0.606 [0.580, 0.633]
judge_cheap_disg	0.584 [0.550, 0.618]
judge_cheap_orig	0.633 [0.601, 0.666]

c.align minus judge_cheap_disg: $\Delta = +0.217$ [$+0.175$, $+0.260$] ($p < 0.0001$).

6.5 Experiment 14: Freeze and GG screening

This experiment freezes V0 (c_score_align), screens five variants against it on development set E, and tests the GG checker.²¹

6.5.1 V0 frozen: development baselines

Table 49: V0 frozen development baselines (Iteration 5, Experiment 14).

Score	ALL-strat	LONG-strat	L25	CTRL
V0 (frozen)	0.742	0.743	0.713	0.738
V2 (reliability-weighted)	0.752	0.750	0.718	0.773
V4 (two-channel logistic)	0.758	0.751	0.719	0.826
judge_cheap_disg	0.641	0.624	0.641	0.807

Selection decision: NONE — V0 stands. No variant met the pre-registered selection rule (highest LONG-strat AUROC beating V0 by $\geq +0.015$). The best candidate (V2) gained $+0.0075$ LONG-strat, below the threshold.

²⁰Artifact `gen_art_experiment_13`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-5/experiment-13

²¹Artifact `gen_art_experiment_14`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-5/experiment-14

6.5.2 GG checker

GG development gate g3: **FAIL**. GG strat AUROC on E = 0.696 [0.659, 0.733] vs V0 0.742, $\Delta = -0.046$ [-0.071, -0.021]. GG’s e/d decomposition: $e = 0.081$, $d = 0.629$. GG has very low endorsement but very high divergence: the gloss gate rejects too many legitimate predicate mappings.

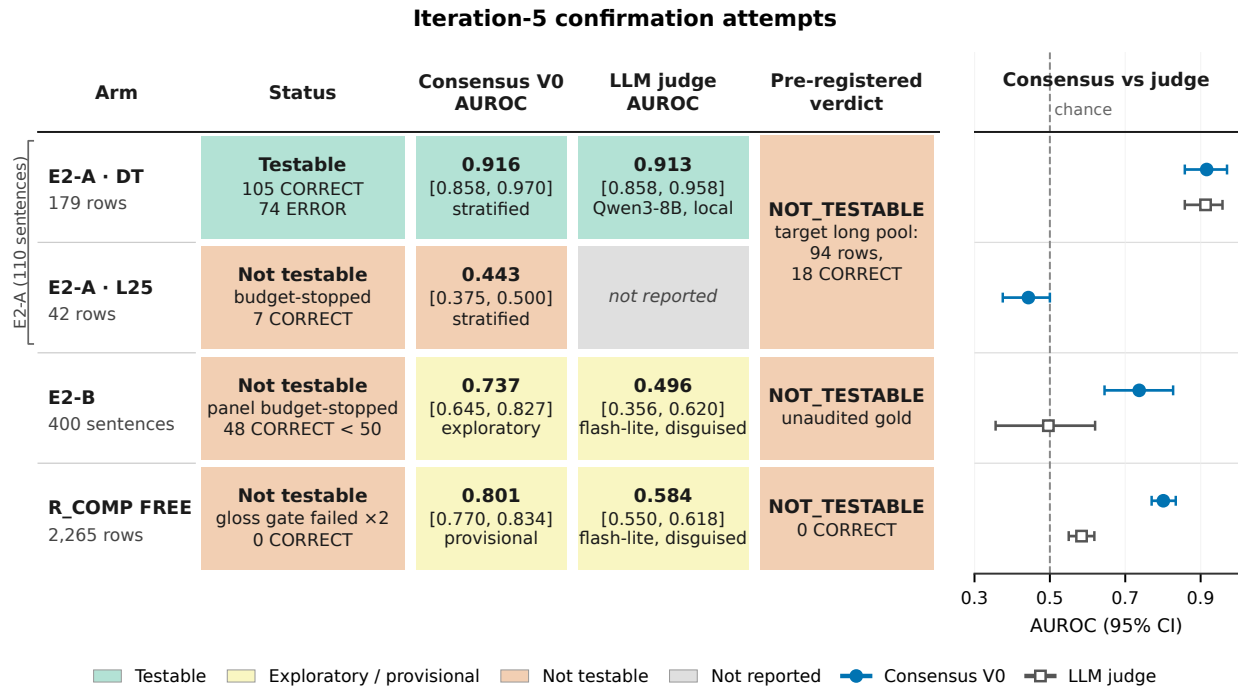


Figure 7: Iteration-5 confirmation attempts and their outcomes. Rows: E2-A (110 sentences), E2-B (400 long sentences), R.COMP FREE (2,265 rows). Columns: status, V0 AUROC [CI], judge AUROC [CI], verdict. No arm reached testability. The only testable cell, E2-A DT, shows high consensus AUROC (0.916) but the local judge performs the same there (0.913).

6.6 Evaluation 4: Corrected record

244 claims were verified against source files. 15 paper-vs-file mismatches were corrected.²²

6.7 Iteration 5 Summary

What iteration 5 established:

1. Confirmation is NOT_TESTABLE across the board. The run-wide budget (\$7.00) was exhausted before any confirmation arm could complete.
2. The DT stratum is a bright spot: V0 reaches strat AUROC 0.916 on ProverQA-derived sentences with trusted gold.

²² Artifact `gen_art_evaluation_4`. Code: https://github.com/ai-inventor-papers/ai-invention-eff060-layered-gold-free-checks-for-logic/tree/fork/run_qck1d05BqxkX/round-5/evaluation-4

3. L25 on E2 is near chance: V0 strat AUROC 0.443 on 42 rows.
4. R_COMP FREE provisional results favour consensus: c_align 0.801 vs judge 0.584.
5. V0 stands: no variant V1–V5 met the selection threshold.
6. GG fails its development gate (strat AUROC 0.696 vs V0 0.742).
7. The rename weakness persists on confirmation data.
8. 15 paper-vs-file mismatches fixed.

6.8 Final dead ends

Table 50: All dead ends across the five iterations.

Dead end	Iter	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557 (below bag-of-words)
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC (provisional)	4	Strat AUROC 0.614, anchoring $e = 0.459$
GG (Gloss-Gated)	5	Gate g3 FAIL: strat AUROC 0.696 vs V0 0.742

Iteration-5 API spend: E2-B \$2.05; R_COMP FREE \$1.97.

7 What We Have Learned

The cross-family consensus metric (`c_score_align`) is the strongest gold-free faithfulness signal found in this investigation. On development set E, it achieves stratified AUROC 0.742, beating the best cheap LLM judge (0.642) by +0.099 [+0.049, +0.146]. On R_COMP SIG (the out-of-scope controlled-vocabulary condition), it reaches 0.954 within-template AUROC with zero error endorsement. Three peer families suffice for 97% of full-pool performance (M4 confirmed: $k_{95} = 3$; by words tercile $k_{95} = 2/3/5$).

The mechanism is clear: errors scatter across z3-equivalence classes while correct translations cluster. The consensus signal degrades with sentence length (NET +0.401 on development E, driven by divergence d), meaning longer sentences need more peers and the metric becomes less reliable precisely where it matters most.

Iteration 5 attempted confirmation on three fronts, and all three were blocked by the run-wide budget (\$7.00). E2-A produced one testable cell: V0 reaches 0.916 on ProverQA-derived sentences with trusted gold, but only 0.443 on the L25 long-sentence stratum (near chance). E2-B’s exploratory results (V0 0.737 vs flash-lite 0.496, $\Delta +0.208$) favour consensus on long sentences with unaudited labels, but the label-coupling diagnostic shows part of the advantage comes from shared gold-equivalent peers. R_COMP FREE’s provisional results show c_align 0.801 vs judge 0.584 ($\Delta +0.217$) under ERROR_CERT vs MAPPED labels, with consensus carrying most of the signal in nesting tests.

No consensus variant V1–V5 improved on V0. V0 stands as the shipped metric.

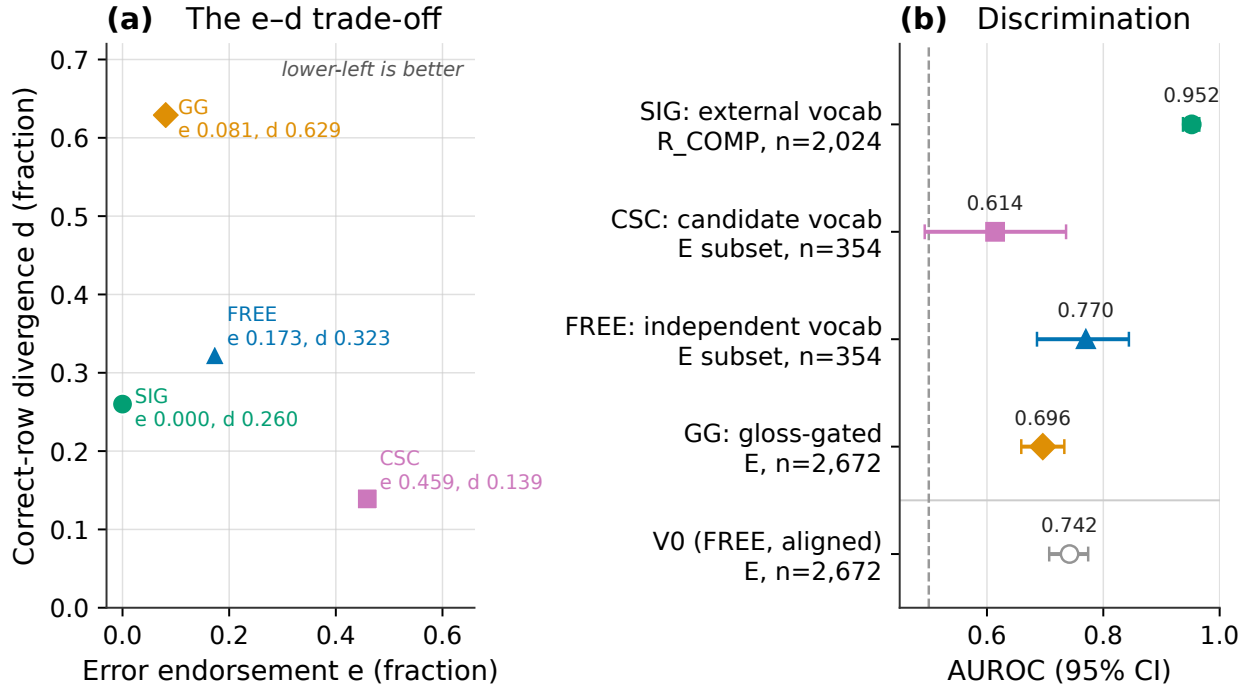


Figure 8: The vocabulary-source trade-off for cross-family consensus. (a) Error endorsement e against correct-row divergence d ; lower-left is better. (b) AUROC per regime. External vocabulary (SIG) reaches $e = 0$ and AUROC 0.952. Candidate vocabulary (CSC) lowers d to 0.139 but anchors peers ($e = 0.459$), giving AUROC 0.614. Independent vocabulary (FREE) keeps $e = 0.173$ at $d = 0.323$, with AUROC 0.770. Gloss-gated filtering (GG) cuts e to 0.081 but raises d to 0.629, failing its development gate (0.696 vs V0 0.742).

The rename tradeoff remains the central open problem. V0 has RENAME_NONCE FA 0.765 on development E and 1.0 on E2. The GG checker was the iteration-5 attempt to solve this; it failed because the gloss gate rejects too many legitimate predicate mappings.

Five limitations bound the current results:

1. No confirmatory hypothesis is confirmed or refuted: every confirmation arm was budget-stopped before reaching testability.
2. All evaluation is on FOLIO and MALLS-derived data; domain transfer is untested.
3. The labels are noisy: the panel majority accuracy on expert-curated errors is 0.727, and the R_COMP FREE audit shows 13–28% label contamination.
4. The consensus advantage does not grow with sentence complexity (M3 INCONCLUSIVE).
5. The rename tradeoff is unsolved: no variant achieves both low rename FA and high error recall.

8 Related Work

NL-FOL datasets and their quality. FOLIO [1] provides 1,435 FOL-annotated NLI problems. MALLS [2] uses GPT-4 to generate and verify NL-FOL pairs at scale. Brunello et al. [4] found that approximately 42% of entries in both datasets contain incorrect FOL formalisations. Logic-LM [10] prompts LLMs for FOL and calls a prover, releasing outputs that we use as our evaluation screen.

FOL closeness metrics. Thatikonda et al. [5] evaluate n-gram, graph, embedding, and LLM-based metrics on controlled FOL perturbations, finding that no single metric is sensitive to all perturbation types.

NL-FOL error taxonomies. Thatikonda et al. [6] propose a verification pipeline with a 28-category error taxonomy. Our typed repair operators follow the same tradition but are derived from z3 equivalence-class differences.

Factuality metrics. FRANK [11] defines a typology of factuality errors in abstractive summarisation. QuestEval [13] uses question generation and answering. QAGS [12] similarly generates questions from a summary. Our L3 questionnaire layer is closest to this family.

Round-trip and self-consistency. Amrollahi et al. [7] propose verbalising a formal statement back to natural language and checking consistency. Li et al. [8] use symbolic equivalence and semantic consistency. Our round-trip baseline achieves only AUROC 0.710, below the LLM judge.

Counter-interpretation methods. CLOVER [9] generates counter-interpretations to test logical validity. Our consensus approach uses multiple independent translations as implicit counter-proposals.

Cross-check consistency. SAC3 [17] detects hallucinations in black-box LLMs by checking consistency across perturbations. Our Δ AUROC of cross-family consensus over single-model self-consistency is +0.140.

Automated alignment for autoformalization. FormalAlign [18] trains a model on both autoformalization and alignment scoring. Our approach uses no learned alignment model.

Cross-model translation confidence. Bayless et al. [19] apply the same score form as `c_score` to NL-to-SMT translation. NoTB [20] uses four LLM families for requirements-to-temporal-logic translation. GenV [21] trains a single verifier for NL-to-FOL, achieving AUROC 0.961 on z3-reference labels but only 0.679 under panel-intent labels.

Self-consistency for formal translation. SCP-NL2TL [22] evaluates single-model self-consistency for NL-to-temporal-logic translation. Our k -saturation finding ($k_{95} = 3$) and the cross-family advantage (+0.140 over SC-5) extend this to cross-family consensus.

Predicate alignment. LogicLLaMA [23] uses a greedy binding search. Vossel et al. [24] map predicates by normalised Levenshtein distance. No prior work reports a rename false-alarm rate.

N-version programming. The consensus mechanism is an instance of N-version programming [25]. Chen and Avizienis [25] noted that faulty but identical results may outvote correct results—precisely the endorsement failure mode we measure ($e = 0.124$ on E, 0.459 under CSC). Eckhardt and Lee [26] predict that coincident failures rise with input difficulty.

References

1. Han et al. (2022). FOLIO: Natural Language Reasoning with First-Order Logic. EMNLP.
2. Yang et al. (2023). MALLS: Multi-Agent Annotated NL-FOL Benchmarks. arXiv:2305.13252.
3. Brunello et al. (2026). Fixing FOLIO and MALLS. arXiv:2606.02837.
4. Brunello et al. (2025). Do LLMs Really Struggle at NL-FOL Translation? AAAI 2026.
5. Thatikonda et al. (2025). Assessing the Sensitivity and Alignment of FOL Closeness Metrics. EMNLP Findings.
6. Thatikonda et al. (2024). Strategies for Improving NL-to-FOL Translation with LLMs. arXiv:2409.16461.
7. Amrollahi et al. (2026). Faithful Autoformalization via Roundtrip Verification and Repair. arXiv:2604.25031.
8. Li et al. (2024). Autoformalize Mathematical Statements by Symbolic Equivalence and Semantic Consistency. NeurIPS.
9. Ryu et al. (2024). CLOVER: Compositional First-Order Logic Translation and Verification. ICLR.
10. Pan et al. (2023). Logic-LM: Empowering Large Language Models with Symbolic Solvers. EMNLP.
11. Pagnoni et al. (2021). FRANK: A Benchmark for Factuality Metrics. NAACL.
12. Wang et al. (2020). QAGS: Asking and Answering Questions to Evaluate Factual Consistency. ACL Findings.
13. Scialom et al. (2021). QuestEval: Summarization Asks for Fact-based Evaluation. EMNLP.
14. Yanaka et al. (2021). SyGNS: A Systematic Generalization Testbed. ACL Findings.
15. Chen et al. (2021). Udep2Mono. EACL.
16. Kupferman and Vardi (2003). Vacuity Detection in Temporal Model Checking. STTT 4(2):224–233.
17. Zhang et al. (2023). SAC3: Reliable Hallucination Detection via Semantic-aware Cross-check Consistency. EMNLP Findings.
18. Lu et al. (2025). FormalAlign: Automated Alignment Evaluation for Autoformalization. ICLR.
19. Bayless et al. (2025). A Neurosymbolic Approach to NL Formalization and Verification. arXiv:2511.09008.
20. Bhatia et al. (2026). NoTB: Multi-LLM Consensus for Reliable NL-to-Temporal-Logic Translation. arXiv:2608.21962.
21. GenV (2026). Generative Verification for NL-to-FOL. arXiv:2609.11085.
22. SCP-NL2TL (2026). Self-Consistency Probing for NL-to-Temporal-Logic. arXiv:2608.05439.
23. Yang et al. (2023). LogicLLaMA. arXiv:2305.15541.
24. Vossel et al. (2025). Advancing NL Formalization to FOL with Fine-tuned LLMs. arXiv:2509.22338.
25. Chen and Avizienis (1978). N-Version Programming. FTCS-8.
26. Eckhardt and Lee (1985). A Theoretical Basis for Multiversion Software Analysis. IEEE TSE 11(12):1511–1517.